ASSIGNMENT 6: Implement Linear Regression Algorithm

1. Aim

To implement the Linear Regression algorithm using Python to predict continuous numerical values and analyse the relationship between independent and dependent variables.

2. Objectives

1. To understand the concept of supervised learning.
2. To study the working principle of Linear Regression.
3. To establish a relationship between input and output variables.
4. To train a Linear Regression model using sample data.
5. To predict numerical values for new inputs.
6. To evaluate model performance using regression metrics.
7. To visualise the relationship between actual values and predicted values.

3. Theory

Linear Regression is a supervised machine-learning algorithm used to predict continuous numerical values. It establishes a mathematical relationship between one or more independent variables and a dependent variable.

For example, Linear Regression can be used to predict student marks based on study hours, house prices based on property features, or sales based on advertising expenditure.

Linear Regression attempts to find the best-fitting straight line through the available observations. The model estimates its parameters by minimising the sum of squared differences between actual values and predicted values.

The equation of simple Linear Regression is:

$$y=a+bx$$

Where:

y = predicted dependent variable.

x = independent variable.

a = intercept.

b = slope of the line.

The intercept represents the predicted value of y when x=0 . The slope indicates the expected change in the predicted output when the input increases by one unit.

Three important evaluation metrics are:

1. Mean Absolute Error (MAE): Measures the average absolute difference between actual and predicted values.

2. Root Mean Squared Error (RMSE): Measures prediction error while giving greater weight to larger errors.

3. R-squared: Measures how well the model explains variation in the target compared with a baseline that predicts the target mean.

Linear Regression is simple and interpretable, but its effectiveness depends on whether a linear relationship reasonably represents the data.

4. Algorithm

1. Start the program.

2. Import NumPy, Matplotlib, and the required Scikit-learn modules.

3. Create sample data containing study hours and marks.

4. Separate the independent variable and dependent variable.

5. Create a Linear Regression model.

6. Train the model using the sample data.

7. Predict the output values.

8. Calculate MAE, RMSE, and R-squared.

9. Display the results.

10. Plot the actual data points and the regression line.

11. Stop the program.

5. Explanation of the Code

1. `numpy` is used to create and manage numerical arrays.

2. `matplotlib.pyplot` is used to display the graph.

3. `LinearRegression()` creates the regression model.

4. `model.fit(X, y)` trains the model using study hours and marks.

5. `model.predict(X)` predicts marks for the input observations.

6. `model.intercept_` returns the estimated intercept.

7. `model.coef_` returns the estimated slope.

8.mean_absolute_error() calculates MAE.

9. `mean_squared_error()` calculates the mean squared error, whose square root gives RMSE.

10. `r2_score()` calculates R-squared.

11. `plt.plot()` draws the fitted regression line.

6. Applications

1. Predicting examination marks from study hours.
2. Estimating sales from advertising expenditure.
3. Predicting house prices from numerical features.
4. Analysing product demand.
5. Forecasting business trends.
6. Studying relationships between scientific measurements.

7. Advantages

1. Easy to understand and implement.
2. Requires relatively little computational power.
3. Provides interpretable coefficients.
4. Useful for identifying linear relationships.

8. Limitations

1. It may not represent complex nonlinear relationships.
2. Outliers can affect the fitted line.

3. Predictions outside the observed data range may be unreliable.

4. Performance depends on the quality of the data.

5. A relationship in the data does not necessarily imply causation.

9. Conclusion

Linear Regression was implemented using Python to study the relationship between study hours and examination marks. The model was trained to estimate the output values, and its performance was measured using MAE, RMSE, and R-squared. A graph was also plotted to visualise the actual observations and the regression line.